INCH-POUND

A-A-54657
29 May 1992

COMMERCIAL ITEM DESCRIPTION

CURING SYSTEM, DENTAL, VISIBLE LIGHT

The General Services Administration has authorized the use of this Commercial Item Description.

This commercial item description covers the complete visible light curing system with the component parts as indicated.

Salient characteristics:

Light Guide, 8mm
Light Guide, 3mm
Bracket Kit, Wall Mounting
Light Shield, Dental
Lamp, Incandescent

General. The curing system shall be composed of a control unit/power supply, a light gun/handpiece, a 3mm (nominal) light guide, an 8mm (nominal) light guide, a lamp, a bracket kit, and a light shield as specified herein. The light gun shall, when not in use, mount in a cradle-type holder located on the top, front portion of the control unit.

Material. The control unit and light gun housings shall be fabricated of high impact, high temperature, engineered thermoplastic resin. The light transmitting material of the light guides shall be coherent low-loss, drawn and fused, glass fiber optic rod. The light shield shall be fabricated of a polycarbonate resin.

Style, design and dimensions. The curing system shall require an input of 110VAC ± 10%, 50/60 Hz, 1.4A. The gun cord shall be a retractile type, flexible stranded, 4 conductor cable, having an uncoiled length of not less than 8.5 feet, and shall be strain relieved on both ends. The power cord shall be a 3 conductor cable having a minimum length of 6 feet, with a hospital grade male plug at one end, and a female receptacle at the other end.

AMSC N/A FSC 6520

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Control unit. The control unit shall house the electronics necessary to operate, control the timing of, and regulate the blower motor, and beeper operation and timing. Voltage to the blower shall be regulated to an 11.0V to 12.1V operating range.

Light gun. The light gun shall house a blower system which pulls air that has been heated by the lamp through rear openings in the gun housing.

The light gun shall also contain a dual filter system which only permits light emitted from the lamp which is in the 400nm to 525nm range to pass through. Light of all wavelengths outside this range shall be reflected through openings in the top of the gun housing. The blower motor shall be rated at 12VDC, 120mA, 800 hours life. The light gun housing shall be of two piece construction such that when the light gun is fully assembled, the two pieces shall provide a tight fit which precludes the collection of debris.

Light guides. The visible light dental curing system shall contain one 3 $\pm$ 0.5mm diameter (over glass portion only) fiber optic light guide and one 8 $\pm$ 0.5mm diameter (over glass portion only) fiber optic light guide. Each light guide shall be bent in an arc, the included angle of which, is between 105^0 and 130^0. The straight ends of each light guide shall be permanently protected from breakage by stainless steel covers.

Each light guide shall be autoclavable and cold sterilizable, as well as completely immersible. The 3mm light guide shall have a yellow filter designed into the light guide housing to reduce glare and obtain optimum light color (green) for dental procedures. All light guides shall have the capability of being rotated freely through 360^0 when inserted in the gun chuck, with no loss of light transmissibility. The interface formed when a light guide is inserted into the gun shall be completely secure to prevent accidental disengagement of guide.

Lamp. The visible light source shall be a tungsten halogen lamp with a power rating of 100W at 12VAC. Voltage provided to the lamp shall be 11.8 $\pm$ 0.2 volts (see note).

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Light shield. The light shield shall be transparent, orange, optically clear, free of surface defects, and shall have a thickness of 0.110 to 0.130 inch. The light shield shall block intense reflected light from the curing system operator's vision, and shall be capable of secure axial rotation when assembled to the front nose cone bushing of the light gun housing.

Bracket kit. The curing system shall contain provisions for table mounting, wall mounting, and for attaching to a 2 inch diameter instrument delivery system pole.

Identification marking. Each curing unit shall be permanently and legibly marked with the name or registered trademark of the manufacturer, model designation, serial number, voltage and frequency.

Instructions. Two manuals providing instructions for how to operate the curing system, lamp and fuse replacement, wall, table, and pole mounting, cleaning, and safety precautions shall be provided with each curing system.

UL Approval. The curing system shall be in accordance with Underwriter's Laboratories (UL) 544.

Finish. The light gun and control unit shall have a smooth finish for ease of cleaning.

Performance. Light output from the system (using 8mm guide) shall be not less than 600mW/cm^2, with the system activated for a period of 20 consecutive seconds. The system shall show no evidence of breakdown after being subjected to the dielectric strength tests as specified in UL 544. The exterior surface temperatures of the light gun handle shall not exceed 60^{0}C (140^{0}F) when running continuously.

Operation. The lamp shall be activated by means of a trigger switch. Lamp activation shall be accompanied by an audible signal. The lamp shall remain on for an operator selected duration of 10, 20, 30, 40, 50 or 60 seconds. Lamp turn-off shall be automatic, and shall be accompanied by an audible signal. The blower unit shall remain on for approximately 4 minutes following lamp turn-off, to dissipate heat and allow lamp cooling.

Workmanship. The unit and all accessories shall be free from defects which may detract from their appearance or impair their serviceability.

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Unit. Each (EA). One curing system, as specified, constitutes one unit. One curing system shall be composed of one control unit, one light gun, one 3mm light guide, one 8mm light guide, power cord, connecting cord, lamp, and one light shield, as specified herein.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after the delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105, at inspection level II, with an AQL of 2.0 (percent defective). Sampling for tests shall be conducted in accordance with MIL-STD-105, at inspection level II, with an acceptance number of zero for all sample sizes. The unit of product for inspection purposes shall be one curing system, as specified herein.

Tests. Test shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

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Vibration test. The curing system shall be secured to a vibrator table in accordance with table manufacturer's instructions. The curing system lamp shall be triggered on. The curing cycle duration shall be set at not less than 20 seconds. The lamp shall remain on for 20 seconds without failure, with the vibrator table control set for 7400 vibrations per minute, with an acceleration force of 6G's.

Performance test. The lamp timing shall be adjusted such that the operator selected time duration for the lamp to remain on, shall be met as a minimum. The maximum allowable time duration shall be less than the selected curing cycle duration plus 2 seconds. The light output shall be not less than 600mW/cm^2, when measured with the 8mm light guide's tip placed 0.5mm (nominal) from a power monitor absorber head (but not touching the head).

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be as specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

National Stock Number	Item Identification
6520-01-296-5760	Curing System, Dental, Visible Light
6520-01-315-3703	Light Guide, 8mm
6520-01-315-3706	Light Guide, 3mm
6520-01-315-3704	Bracket Kit, Wall Mounting
6520-01-315-3705	Light Shield, Dental
6240-01-315-3676	Lamp, Incandescent

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NOTE: The curing system is compatible with the following lamp:

6240-01-315-3676	Lamp, Incandescent

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6520-2524
Location: DC Disk #2
(CURING)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	A-A-54657	92-05-29

3. DOCUMENT TITLE
CURING SYSTEM, DENTAL VISIBLE LIGHT

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Defense Personnel Support Center	(1) Commercial (215)737-2870	(2) AUTOVON 444-2870
c. ADDRESS (Include Zip Code) ATTN: DPSC-MSE 2800 S. 20th Street Philadelphia, PA 19101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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